

Green Energy & Hydrogen Fuel Cells

Powering a Sustainable Future

Introduction to Green Energy

The world faces an urgent need to transition away from fossil fuels. Hydrogen fuel cells have emerged as a cornerstone of clean energy strategy — converting hydrogen directly into electricity with zero harmful emissions, no combustion, and remarkable versatility.

Climate Imperative

Global push for carbon-neutral energy systems to limit warming and meet net-zero targets

Hydrogen's Role

Clean electrochemical conversion of hydrogen to electricity — water vapour the only by-product

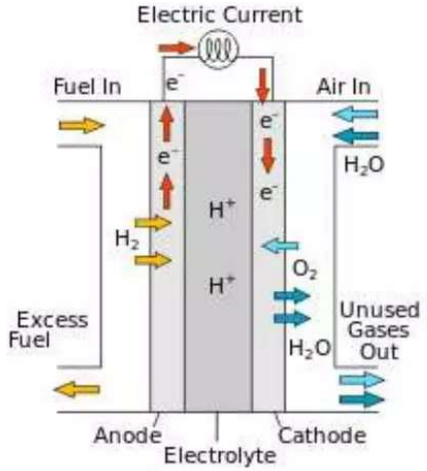
Growing Adoption

Expanding from transport and mobility to stationary power, industry, and grid balancing



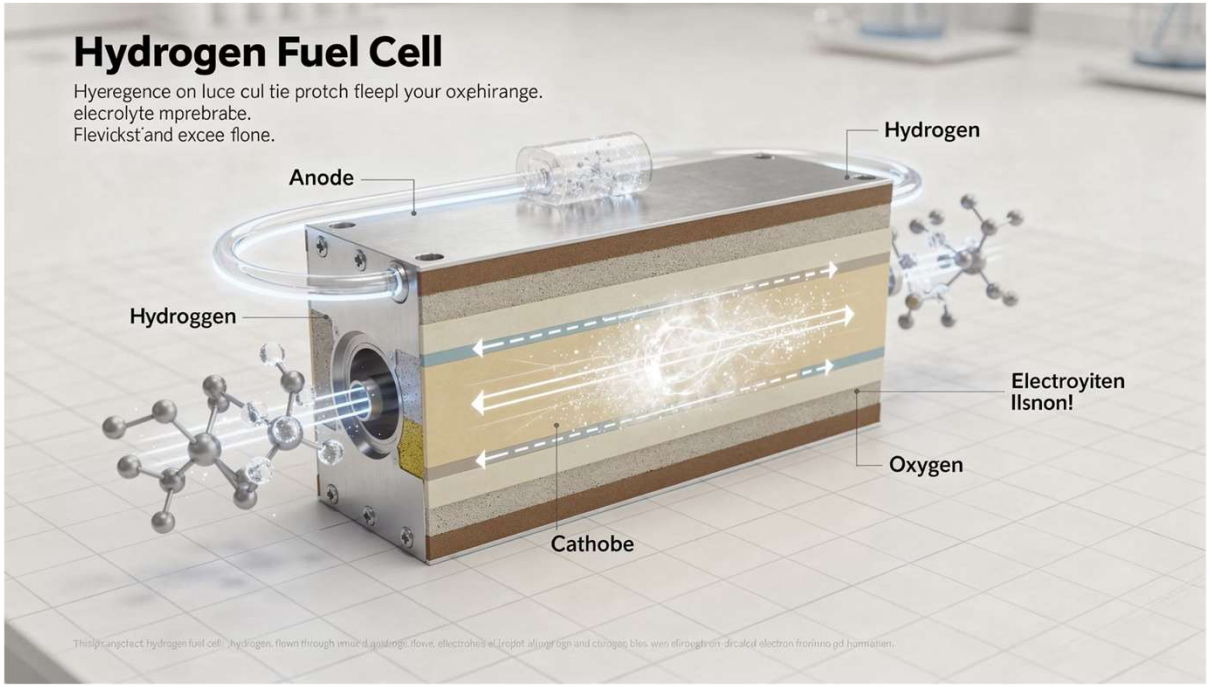
Fuel cell

- A fuel cell produces electricity through a chemical reaction, but without combustion.
- A fuel cell is an electrochemical device that combines **hydrogen** and **oxygen** to produce electricity, with water and heat as its by-product.
- In its simplest form, a single fuel cell consists of two electrodes - an anode and a cathode - with an electrolyte between them.



Fuel cell working

- The reactions that produce electricity happen at the electrodes.
- Every fuel cell has two electrodes, one positive, called the anode, and one negative, called the cathode.
- Fuel goes to the anode side, while oxygen (or just air) goes to the cathode side.
- A chemical catalyst speeds up the reactions here.



CORE SCIENCE

Operating Principles

A hydrogen fuel cell generates electricity through a controlled electrochemical reaction — not combustion. The process is clean, continuous, and highly efficient.

1 Hydrogen to the Anode

H_2 molecules are split into protons and electrons at the anode catalyst layer

2 Electron Flow = Power

Electrons travel through an external circuit, generating usable electrical current

3 Water as By-product

Protons combine with oxygen at the cathode, producing only water vapour and heat

Classification of Fuel Cells

Fuel cells are categorised by their electrolyte type, which determines operating temperature, efficiency, and best application. Each type offers distinct advantages for specific energy needs.

PEM

Polymer Electrolyte Membrane (Eg. Proton Exchange Membrane Fuel Cell) — Low temperature (60–80°C). Fast start-up; ideal for vehicles and portable power. Dominant in hydrogen applications.

SOFC

Solid Oxide — High temperature (600–1000°C). High efficiency; best suited for large stationary power generation.

AFC

Alkaline Fuel Cell — Uses potassium hydroxide electrolyte. High efficiency; originally developed for space missions.

PAFC / MCFC

Phosphoric Acid & Molten Carbonate — Mid-to-high temperature systems for commercial and utility-scale stationary power.

Key insight: PEM fuel cells dominate hydrogen applications due to their versatility, compact design, and high power density at low operating temperatures.

PEM Fuel Cell

Hydrogen is oxidized at the anode of a fuel cell to initiate the production of electricity by splitting into protons and electrons.

This occurs because a catalyst, typically platinum, breaks the bond between hydrogen atoms.

The electrons are forced through an external circuit, while protons move through the electrolyte.

At Anode: $2H_2 \longrightarrow 4H^+ + 4e^-$

At Cathode: $4H^+ + 4e^- + O_2 \longrightarrow 2\ H_2O$

Overall cell reaction: $2H_2 + O_2 \longrightarrow 2H_2O$

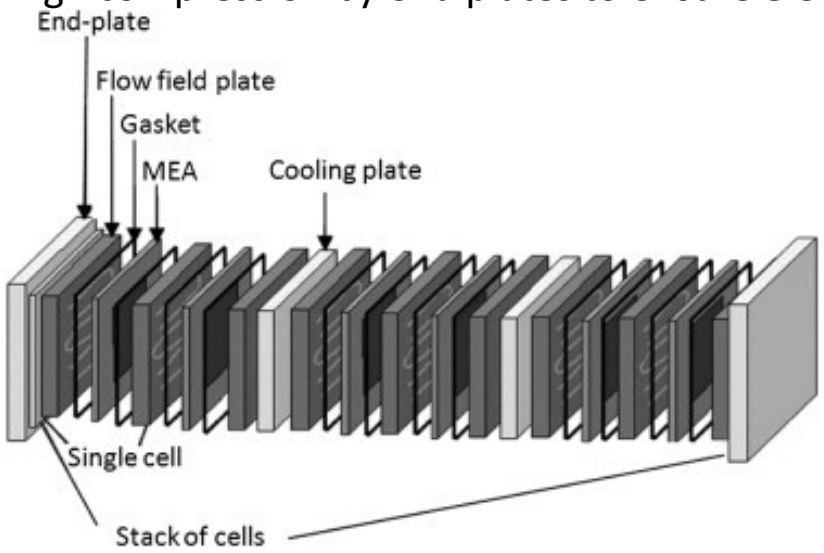
Fuel Cell Stacking

A single cell typically produces less than 1 volt. Stacking in series multiplies the voltage to useful levels for applications.

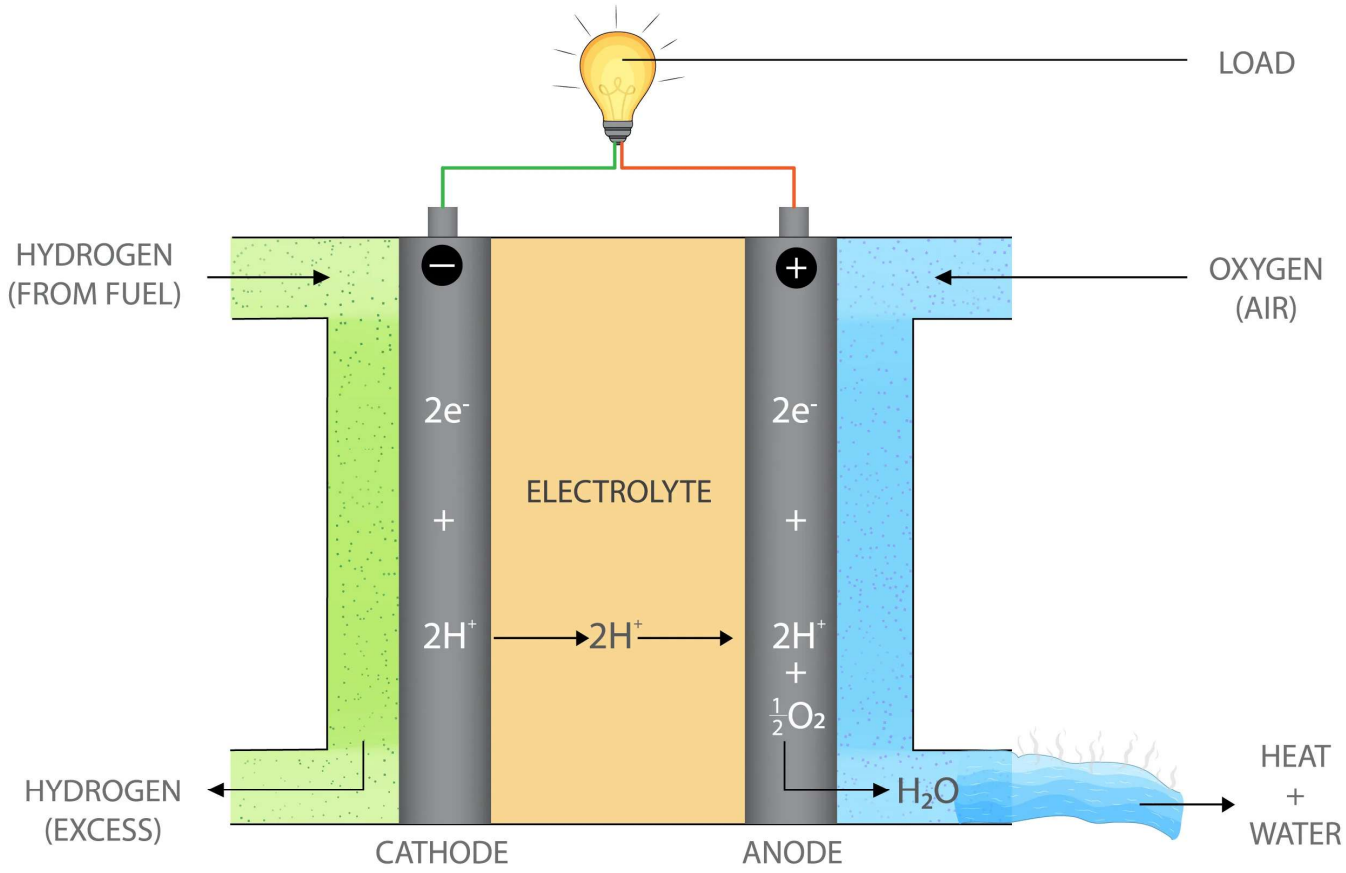
Fuel cell stacking is the process of connecting multiple individual, low-voltage electrochemical cells (**membrane electrode assemblies (MEAs)**)in series (and sometimes parallel) to create a high-voltage, high-power module.

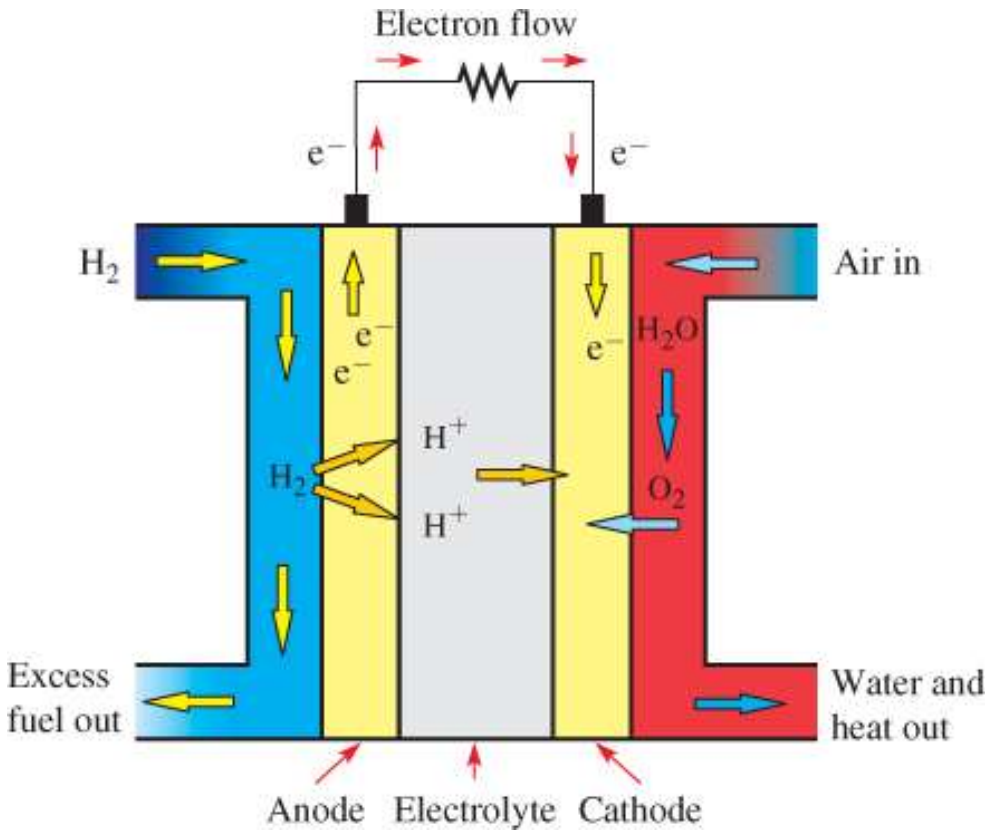
Stacking allows for compact, scalable power generation, with the number of cells customized for applications ranging from drones to heavy-duty trucks or power plants.

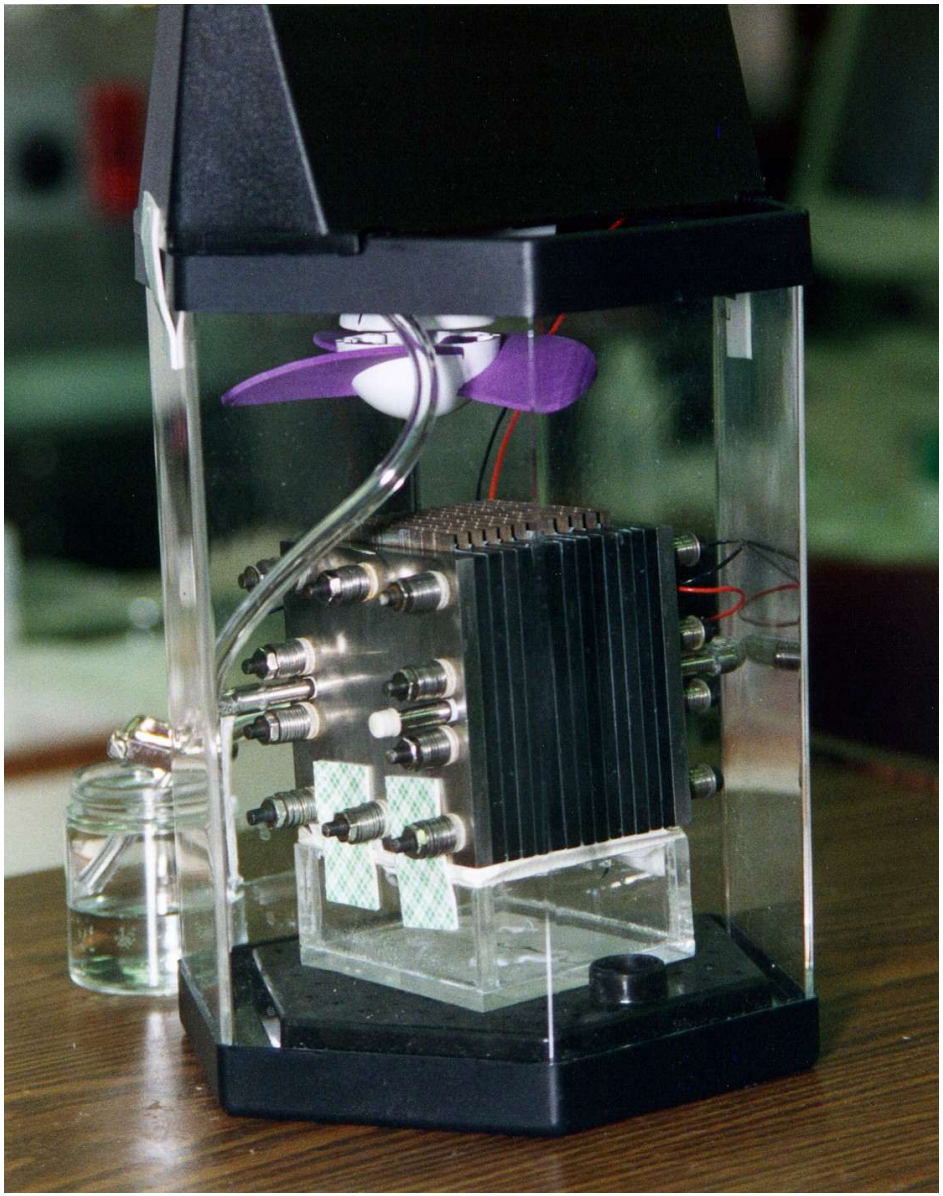
The cells are held together under high compression by end plates to ensure electrical contact and minimize resistance.



FUEL CELL

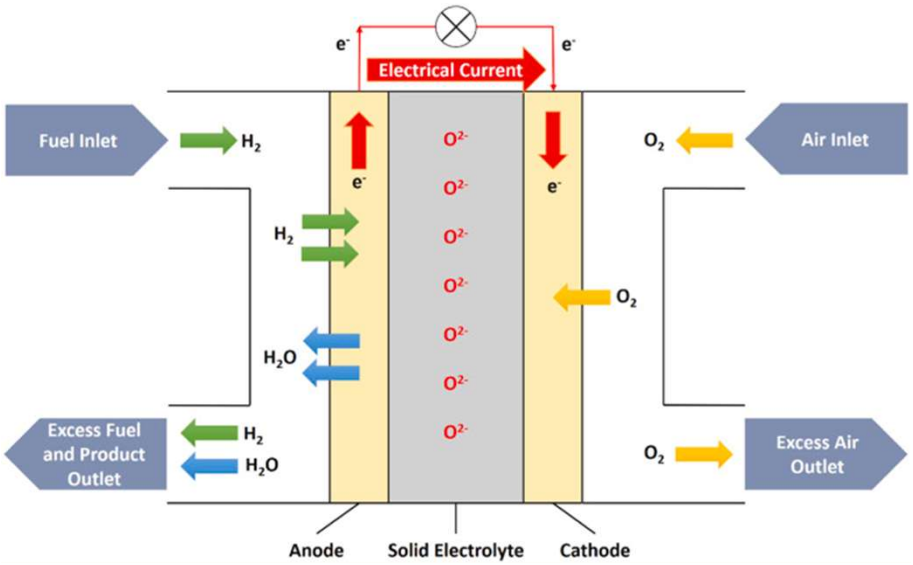

LOAD
HYDROGEN
(FROM FUEL)
OXYGEN
(AIR)
2e-
+
2H+
ELECTROLYTE
2H+
2e-
+
2H+
+
½O2
H2O
HYDROGEN
(EXCESS)
HEAT
+
WATER
CATHODE
ANODE


Electron flow
e⁻
e⁻
H₂
Air in
e⁻
e⁻
e⁻
H₂O
H⁺
H₂
O₂
H⁺
Excess
fuel out
Water and
heat out
Anode
Electrolyte
Cathode



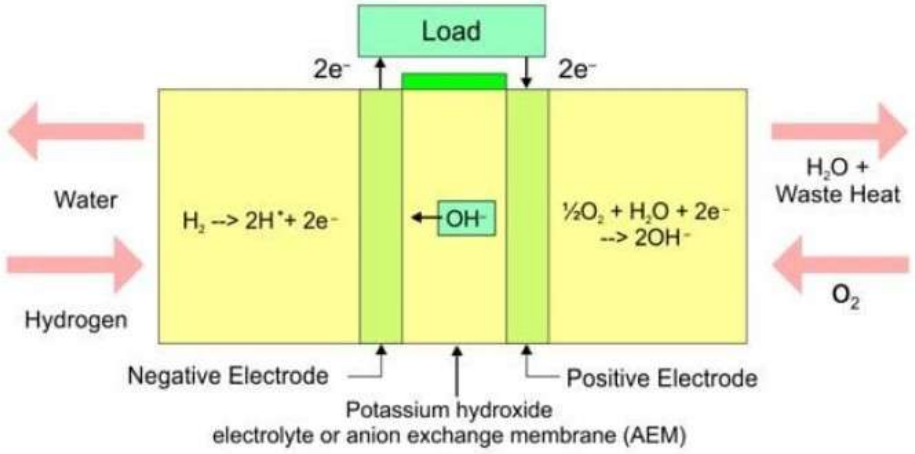
Solid oxide fuel cells (SOFCs)

- **Electrolyte:** A dense, non-porous yttria-stabilized zirconia (YSZ) layer that conducts O^{2-} ions from the cathode to the anode while acting as an electronic insulator.
- **Anode (Fuel Electrode):** A porous ceramic-metal composite (cermet), typically Nickel-YSZ, where fuel oxidation occurs. It is often used as the supporting structure for the cell.
- **Cathode (Air Electrode):** A porous, electronically conducting ceramic, such as Strontium-doped Lanthanum Manganite (LSM), where oxygen reduction occurs.



Alkaline Fuel Cell (AFC)

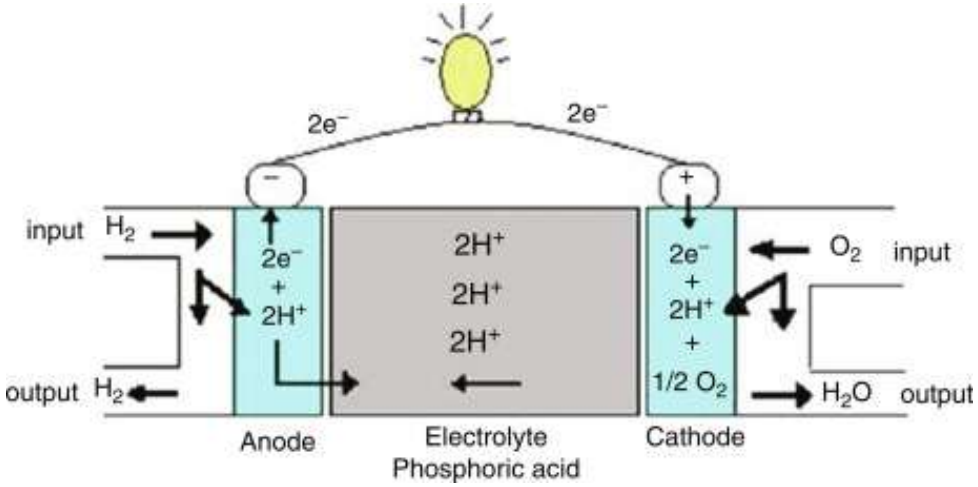
- **Anode:** Typically porous nickel, often coated with a metal catalyst (e.g., Platinum) to catalyze the oxidation of hydrogen.
- **Cathode:** Porous nickel or silver is commonly used, acting as a catalyst for oxygen reduction.
- **Electrolyte:** A concentrated solution (35%–40%) of aqueous Potassium Hydroxide (KOH) is used. This liquid is often held within an asbestos or similar matrix.



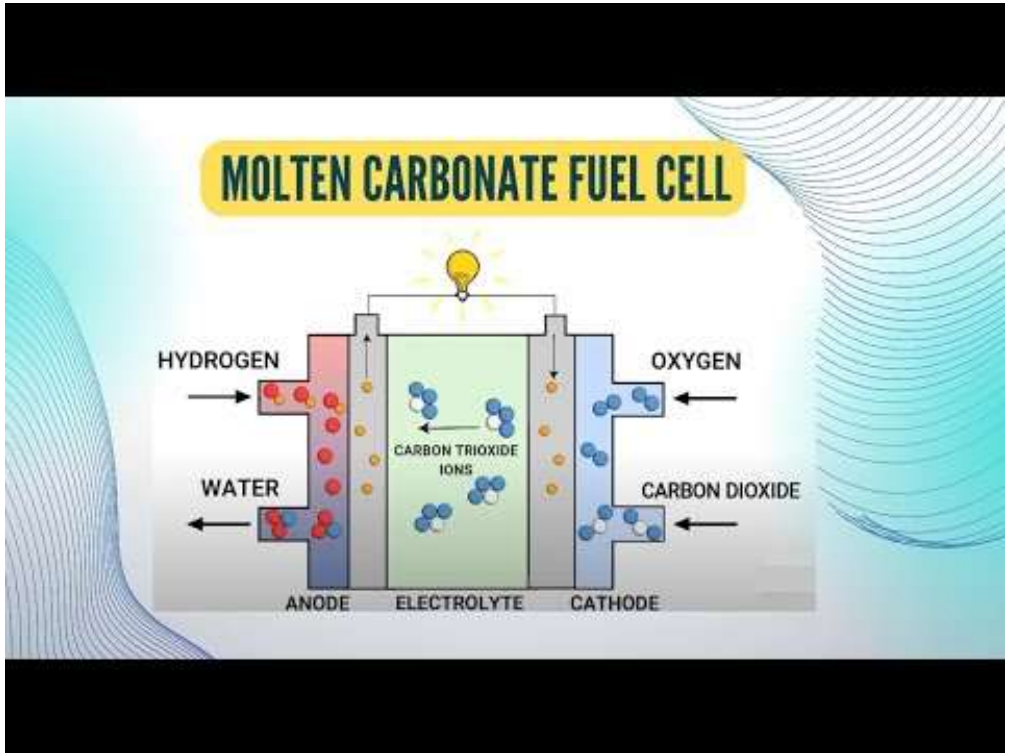
Phosphoric acid fuel cells (PAFCs)

Phosphoric acid fuel cells (PAFCs) consist of a sandwich structure containing a liquid phosphoric acid electrolyte immobilized in a silicon carbide matrix, placed between two porous carbon electrodes (anode and cathode) containing a platinum catalyst

In materials science, a **matrix** is the bulk material, the continuous phase that holds reinforcing particles or fibers together (like a glue) to form a composite.

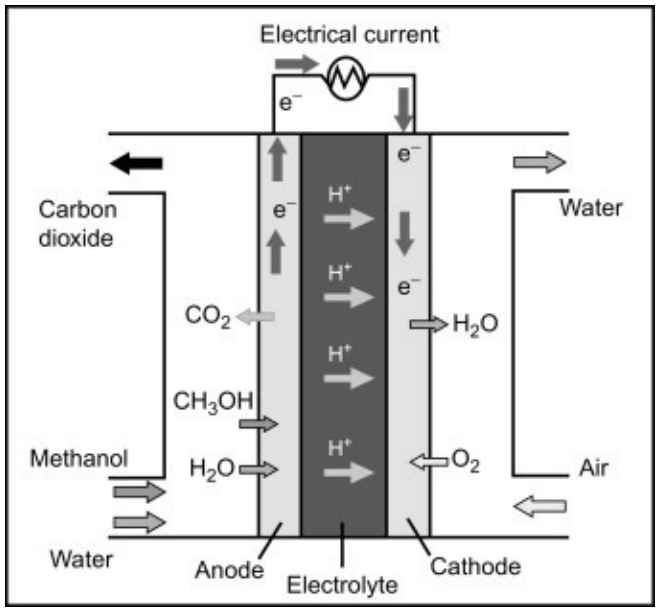


Molten Carbonate Fuel Cells (MCFCs)



Molten Carbonate Fuel Cells (MCFCs) are high-temperature (600–700°C) fuel cells that use a molten salt electrolyte (lithium/potassium or lithium/sodium carbonate) to generate electricity directly from hydrogen or hydrocarbons, such as natural gas, without external reforming.

Direct Methanol Fuel Cell (DMFC)



A Direct Methanol Fuel Cell (DMFC) produces electricity by directly oxidizing liquid methanol at the anode, typically using a Platinum-Ruthenium catalyst, while oxygen is reduced at the cathode.

It operates by using a proton exchange membrane to transport ions, generating electricity, heat, and water while emitting carbon dioxide

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Fuel Cell	Operating Temperature (°C)	Power (kW)	Efficiency (%)	Application
PEM (Polymer Electrolyte Membrane)	60–110	0.01–250	40%–55%	Portable, Mobile, Low power generation.
AFC (Alkaline Fuel Cell)	70–130	0.1–50	50%–70%	Mobile, space, military.
PAFC (Phosphoric Acid Fuel Cell)	175–210	50–1000	40%–45%	Medium to large scale power generation and CHP (Combined Heat and Power).
MCFC (Molten Carbonate Fuel Cell)	550–650	200–100,000	50%–60%	Large scale power generation.
SOFC (Solid Oxide Fuel Cell)	500–1000	0.5–2000	40%–72%	Vehicle auxiliary power units, medium to large scale power generation and CHP, off-grid power and micro CHP.
DMFC (Direct Methanol Fuel Cell)	70–130	0.001–100	40%	Mobile, portable.

Type of fuel cell	Applications	Core temp. efficiency	Advantages	Limitations
Alkaline (AFC)	Space, military, submarines, transport	90–100°C; 60% efficient	Low parts and, operation costs; no compressor; fast cathode kinetics	Large size; sensitive to hydrogen and oxygen impurities
Molten Carbonate (MCFC)	Large power generation	600–700°C; 45–50% efficient	High efficiency, flexible to fuel, co-generation	High heat causes corrosion, long startup, short life
Phosphoric Acid (PAFC)	Medium to large power generation	150–200°C; 40% efficient	Good tolerance to fuel impurities; co-generation	Low efficiency; limited-service life; expensive catalyst
Proton Exchange Membrane (PEMFC)	Portable, stationary and automotive	50–100°C; 80°C typical; 35–60% efficient	Compact design, long operating life, quick start-up, well developed	Expensive catalyst; needs chemical grade fuel; complex heat and water control
Solid Oxide (SOFC)	Medium to large power generation	700–1000°C; 60% efficient	Lenient to fuels; can use natural gas, highly efficient	High heat causes corrosion, long startup, short life

Disadvantages of fuel cell

- Main disadvantage of fuel cell is its high initial cost.
- Fuel cell has low service life.
- If hydrocarbon is the energy source in fuel cell then it emits carbon dioxide and no longer a green source.
- It is comparetively expensive.
- It is difficult to handle pure hydrogen.

Advantages of hydrogen fuel

Environmental Friendliness: Hydrogen fuel cells produce zero carbon emissions, emitting only water vapor and warm air, which improves air quality and helps combat climate change.

High Efficiency: Fuel cells are highly efficient, converting a higher percentage of chemical energy directly into electricity compared to traditional internal combustion engines.

Versatility and Security: Hydrogen can be produced domestically from various sources, including water, natural gas, and biomass, reducing dependence on foreign oil.

Quick Refuelling and High Performance: Hydrogen vehicles have fast refuelling times similar to traditional gasoline, and they offer high performance with low noise levels.

Energy Storage: It serves as a sustainable energy carrier, allowing for the storage of renewable energy from sources like wind and solar for later use.

Durability: Hydrogen fuel cells are known for being long-lasting, with some systems operating for over 20,000 hours with minimal maintenance.

Zero energy Concept

Zero energy typically refers to a building, system, or process that consumes no net energy from the grid, balancing its consumption with on-site renewable energy generation.

These net-zero energy (NZE) structures use high-efficiency insulation, windows, and appliances to minimize energy usage, relying on solar or wind power to meet remaining demands, resulting in near-zero carbon emission

The United States Department of Energy defines the zero-energy community as *"an energy-efficient community where, on a source energy basis, the actual annual delivered energy is less than or equal to the on-site renewable exported energy*

Contributes less greenhouse gas than a normal system/building



Zero-Emission Concept & Benefits of Hydrogen Energy

Zero Greenhouse Gases

No CO_2, NO_x, or particulates at point of use — only water vapour emitted

High Efficiency

50–60% electrical efficiency; up to 80% in combined heat and power (CHP) systems

Most Abundant Element

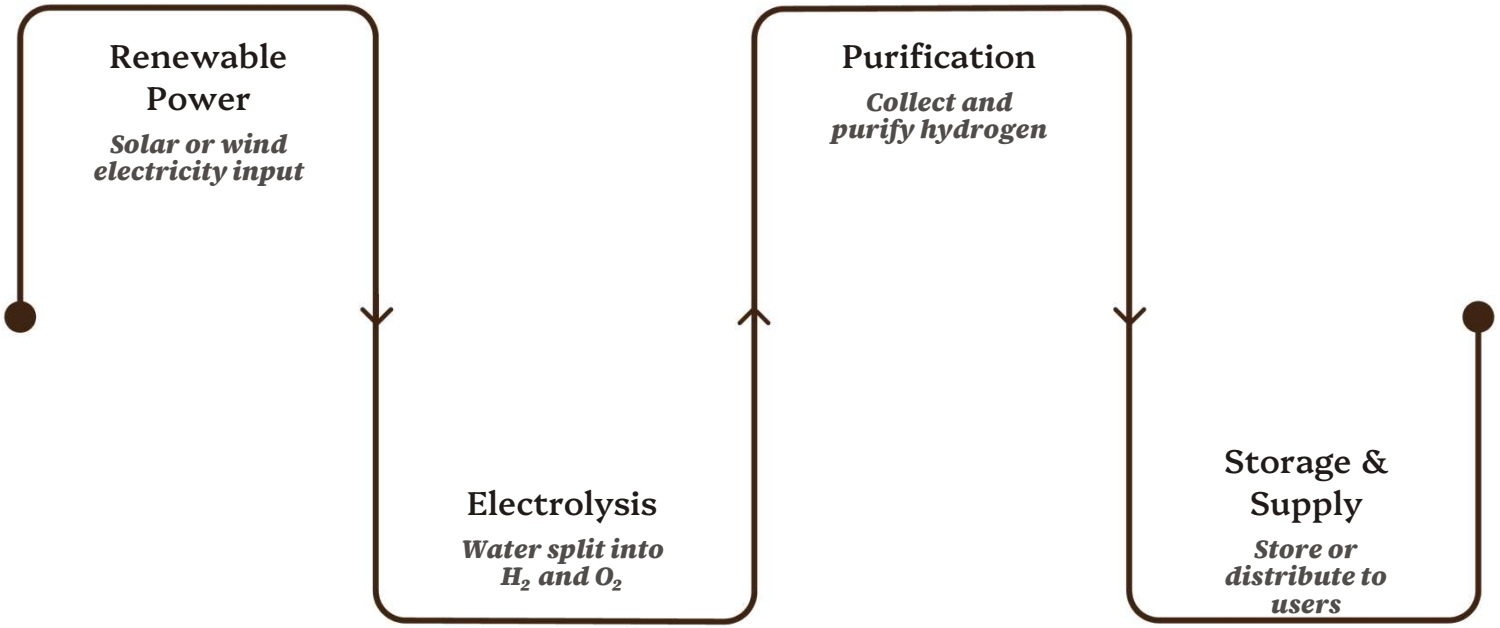
Hydrogen is the universe's most abundant element — renewable when produced sustainably

Versatile & Scalable

Quiet, reliable power for vehicles, buildings, remote sites, and grid-scale storage

Hydrogen Production: The Electrolysis Method

Electrolysis is the most promising pathway to green hydrogen — using electricity to split water (H_2O) into hydrogen and oxygen. When powered by renewables, it achieves a zero-carbon footprint.



Alkaline Electrolyser

Mature, low-cost technology using liquid alkaline solution; suitable for large-scale production

PEM Electrolyser

High efficiency and fast response; ideal for coupling with variable renewable energy sources

Solid Oxide Electrolyser

Operates at high temperatures; highest efficiency potential using waste heat from industry

Electrolysis hydrogen production

Electrolysis hydrogen production splits water into hydrogen and oxygen using electricity, producing zero emissions when powered by renewables.

The three main technologies are Alkaline (low cost/mature), Proton Exchange Membrane (PEM) (fast response/high purity), and Solid Oxide (high efficiency/high temperature).

Key Electrolysis Technologies

Alkaline Water Electrolysis (AEL): The oldest, most mature technology. It uses a liquid alkaline solution (potassium hydroxide) and nickel-based electrodes. Suitable for large-scale, stationary applications, but has a slower response to variable renewable energy.

Proton Exchange Membrane Electrolysis (PEM): Uses a solid polymer electrolyte, allowing faster ramp-up times and high efficiency. PEM is compact and ideal for fluctuating, intermittent renewable energy sources like wind and solar.

Solid Oxide Electrolyzer Cells (SOEC): Operate at high temperatures, allowing for higher efficiency and lower electricity consumption by using waste heat. It is highly efficient but currently in earlier stages of development compared to AEL and PEM.

Performance and Characteristics

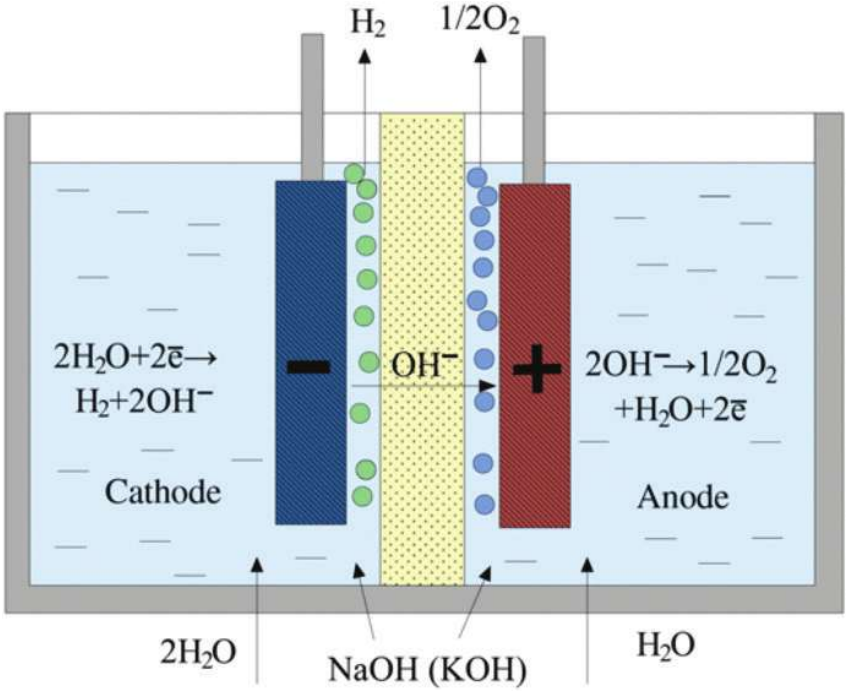
Efficiency: Modern electrolysis processes have electrical efficiencies of 70–80%.

Requirements: Producing 1 kg of hydrogen generally requires 50–55 kWh of electricity.

Advantages: Provides high-purity hydrogen suitable for fuel cells and industrial use.

System Scale: Ranging from small-scale, decentralized units to large-scale, megawatt-level facilities

Alkaline Water Electrolysis (AEL)

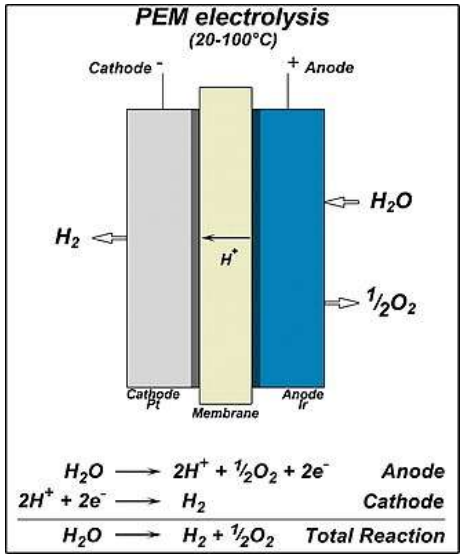


AEL is widely used for industrial-scale hydrogen production

Alkaline Water Electrolysis (AEL) is a mature, cost-effective, and durable technology for producing green hydrogen by splitting water into hydrogen and oxygen using a liquid alkaline electrolyte (typically 25-40 wt% kOH or NaOH)

It operates at 50-90°C, utilizing non-noble metal catalysts (like Nickel), and is suitable for large-scale industrial applications, though it has lower efficiency and dynamic flexibility compared to PEM electrolysis

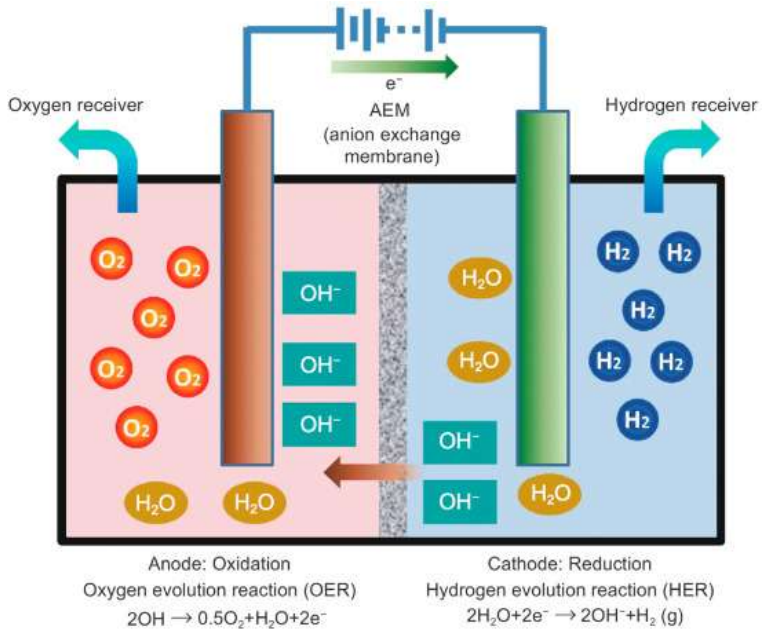
Proton Exchange Membrane Electrolysis (PEM)



It employs a solid polymer electrolyte membrane to conduct protons, allowing for high-pressure operation, fast response times, and high-purity hydrogen production, making it ideal for variable renewable energy sources.

The PEM (often Nafion) conducts protons from the anode to the cathode while acting as an electronic insulator and gas separator.

Solid oxide electrolyzer cells (soec)

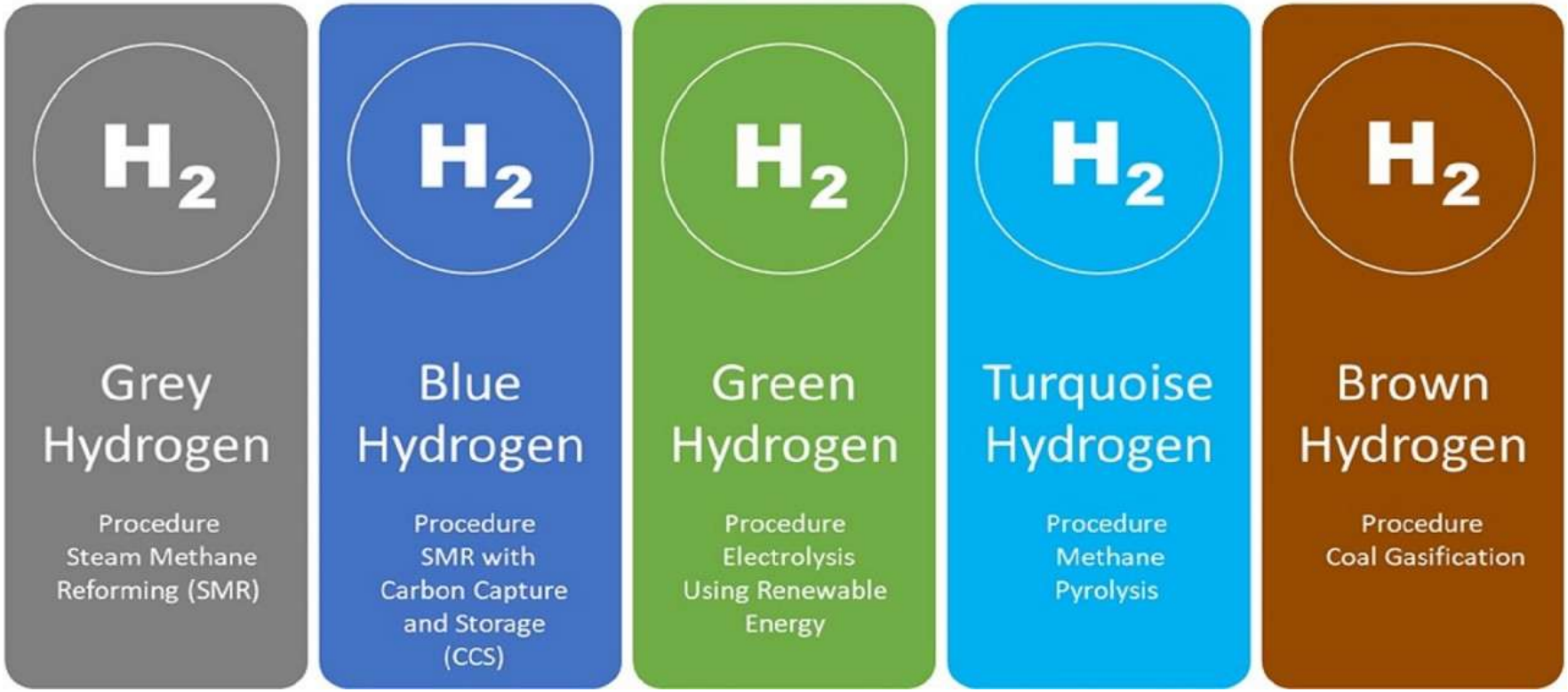


Solid Oxide Electrolyzer Cells (SOEC) are high-temperature electrolysis devices, typically operating at 600–1000°C, that offer 20–30% higher efficiency than low-temperature methods to produce green hydrogen.

Using ceramic electrolytes, they split steam into hydrogen and oxygen, with high efficiency and potential for integrating waste heat

Steam is fed to the cathode, where it is reduced to hydrogen and oxygen ions. These ions travel through a solid ceramic electrolyte to the anode, where they release electrons to form oxygen gas

TYPES OF HYDROGEN



INFRASTRUCTURE

Hydrogen Energy Storage

Hydrogen's low volumetric energy density means storage is a critical engineering challenge. Multiple technologies address different scales and applications, from onboard vehicle tanks to grid-scale energy buffers.

→ Compressed Gas

Stored at 350–700 bar in high-strength tanks. Most common for FCEVs and stationary backup systems

→ Liquid Hydrogen

Cooled to –253°C; high energy density for aerospace and large-scale transport applications

→ Solid & Chemical Storage

Metal hydrides and Liquid Organic Hydrogen Carriers (LOHCs) offer safer, denser alternatives currently under development

Why Storage Matters

Effective hydrogen storage is essential for balancing the intermittent nature of renewable energy — storing surplus solar and wind energy as hydrogen for on-demand use.

700

Bar pressure

Max tank compression for vehicle storage

–253°C

Liquid H_2 temp

Cryogenic liquefaction threshold

Applications of Hydrogen Energy



Transportation

FCEVs deliver 300+ mile range and refuel in ~5 minutes. Buses, trucks, trains, and ferries are rapidly adopting hydrogen for zero-emission mobility.



Stationary Power

Reliable backup and primary power for hospitals, data centres, and remote telecom towers where grid resilience is critical.



Industrial Uses

Green hydrogen replacing coal in steel production, serving as a chemical feedstock, and providing high-temperature industrial heat.



Maritime & Heavy Transport

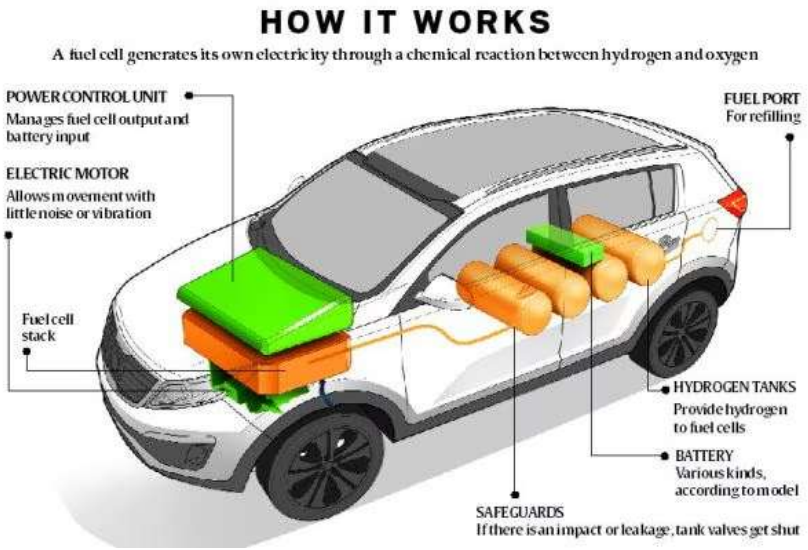
Shipping and heavy-duty freight sectors adopting hydrogen to decarbonise sectors where battery solutions are impractical.

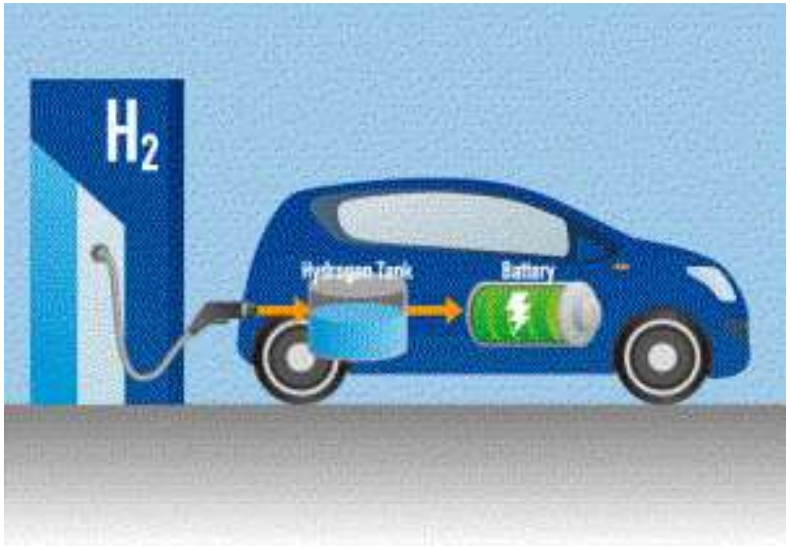
Hydrogen-powered cars, or fuel cell electric vehicles (FCEVs), use hydrogen gas to generate electricity on board, emitting only water vapor.

Leading models include the Toyota Mirai, Hyundai Nexo, and emerging technology like the Hyundai N Vision 74.

They offer quick refuelling (similar to gasoline) and long ranges (300-400+ miles), but face challenges regarding limited fueling infrastructure and high costs

Hydrogen stored in high-pressure tanks combines with oxygen in a fuel cell stack to create electricity, which powers an electric motor.







Toyota Mirai

Challenges & Problems Associated with Hydrogen Energy

Despite its enormous promise, hydrogen energy faces significant technical, economic, and societal barriers that must be addressed to enable widespread deployment.

Infrastructure Gaps

Hydrogen refuelling stations remain sparse globally. Building out pipelines, storage, and dispensing infrastructure requires substantial capital investment and coordinated policy support.

Production Cost

Green hydrogen produced via renewable electrolysis currently costs 3–5× more than grey hydrogen from natural gas. Scale-up and falling electrolyser costs are gradually closing this gap.

Storage & Safety

Hydrogen is highly flammable and requires stringent handling protocols. Energy losses during compression and liquefaction reduce overall system efficiency.

Awareness & Regulation

Public understanding of hydrogen safety is low, and consistent international regulatory frameworks and standards are still developing, slowing commercial deployment.

Conclusion: A Cornerstone of Clean Energy

Hydrogen fuel cells represent one of the most compelling pathways to a zero-carbon energy system — clean, efficient, and adaptable across sectors.

Clean & Efficient

Zero emissions at point of use; high electrical and combined heat efficiency across all scales

Technology Advancing

Electrolysis costs falling; storage technologies maturing; PEM fuel cells improving in durability

Challenges Surmountable

Infrastructure, cost, and safety barriers are real but addressable with investment and policy alignment

Sustainable Future

Hydrogen and fuel cells, combined with renewables, can drive a full transition to a net-zero world

> ***"Hydrogen is not just a fuel — it is the energy carrier that can connect renewable electricity to every sector of our economy."***

